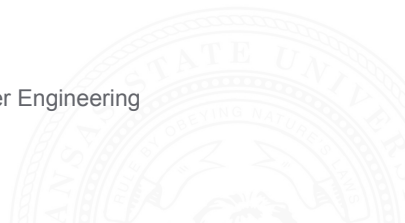


Three-Phase Transformers I

Windings, configurations, and phase shift

Buxin She

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Lecture outline

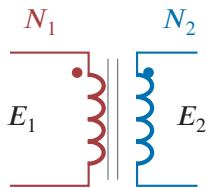
- 1 Winding count, phase count, and physical construction.
- 2 Coil connections for four three-phase banks.
- 3 Terminal relations and phase shift, followed by a numerical bank example.

Follow each connection from its coils to its terminal phasors.

PART 1

Windings and phases

Winding and magnetic coupling



two isolated winding circuits

Both coils share core flux Φ .

$$\frac{E_1}{N_1} = \frac{E_2}{N_2}$$

Winding and voltage symbols

A winding is a coil of insulated conductor.

N_1, N_2 : turns in windings 1 and 2.

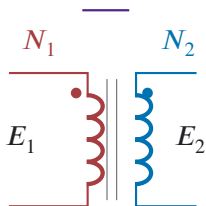
E_1, E_2 : induced RMS voltage phasors.

V_{xy} : RMS terminal-voltage phasor; x relative to y .

For an ideal winding, $E = V$ at its terminals with the same polarity reference.

Two-winding and three-winding

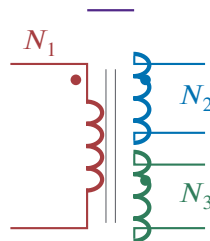
Two-winding



two isolated winding circuits

$$E_1/N_1 = E_2/N_2$$

Three-winding



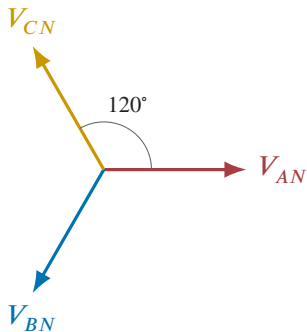
three isolated winding circuits

$$E_1/N_1 = E_2/N_2 = E_3/N_3$$

Three-winding adds a third voltage circuit coupled to the same flux.

Source: Glover et al., sixth edition (2017), Sec. 3.6, Fig. 3.20

Phase and phase systems



A, B, C : supply-phase terminals.
 N : common phase-voltage reference.

Voltage notation

$v(t)$: instantaneous voltage.

V : RMS magnitude; ϕ : phase angle.

f : frequency; $\omega = 2\pi f$: angular frequency.

$$v(t) = \sqrt{2}V \cos(\omega t + \phi)$$

Balanced positive sequence:

$$V_{AN} = V \angle 0^\circ$$

$$V_{BN} = V \angle (-120^\circ)$$

$$V_{CN} = V \angle 120^\circ$$

Use E for induced winding voltage and V_{xy} for voltage between terminals.

Winding count and phase count

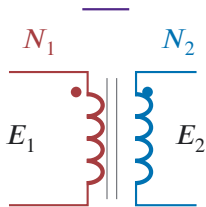
	Single-phase	Three-phase
Two-winding	2 coil circuits	2 winding sets; 6 phase coils
Three-winding	3 coil circuits	3 winding sets; 9 phase coils

For the conventional constructions shown here, each three-phase winding set contains one coil circuit per phase.

A three-phase two-winding transformer has two winding sets, each with three phase coils.

Single-phase and three-phase

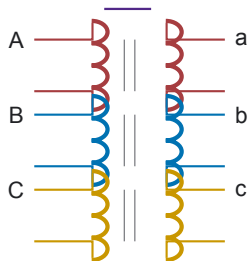
Single-phase



two isolated winding circuits

One alternating voltage system.

Three-phase

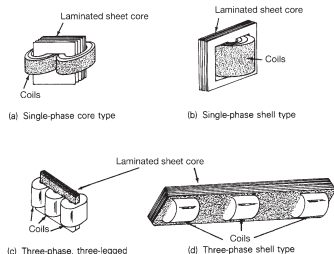


A/a , B/b , C/c : matching phase-coil pairs.

Supply phases are separated by 120° .

Three-phase operation requires three corresponding phase winding pairs.

Bank and common core



Two physical implementations

A bank uses three separate single-phase transformers.

A three-phase unit places the winding sets on a common three-phase core.

Both can realize the same terminal connection.

Core construction and external Y/ Δ connection are separate choices.

Source: Glover et al., sixth edition (2017), Fig. 3.15

PART 2

Connections and phase shift

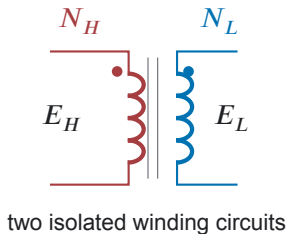
Terminal and phase conventions

- 1 H and L now identify high- and low-voltage sides; their coil quantities use H/L subscripts.
- 2 H phases are A, B, C ; L phases are a, b, c .
- 3 Terminal marks: H_1, H_2, H_3 on H; X_1, X_2, X_3 on L.
- 4 Positive sequence: $A \rightarrow B \rightarrow C$; RMS phasors.
- 5 Compare the corresponding line voltages $V_{AB,H}$ and $V_{ab,L}$.

$$a = \frac{N_H}{N_L} = \frac{|E_H|}{|E_L|}, \quad \delta_{LH} = \angle V_{ab,L} - \angle V_{AB,H}$$

Winding turns ratio a and signed terminal displacement δ are different quantities.

Dots and matching coils



Measure each winding voltage from its dotted to its undotted end.

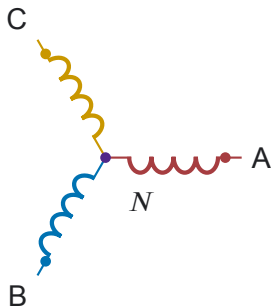
$$E_H = aE_L$$

Same-color coils are magnetically paired.

Wire-junction dots and winding-polarity dots have different roles.

A dot transfers coil polarity; the external terminal joining determines the bank phase shift.

Y: line and coil quantities



Balanced positive sequence

V_{AN} : voltage across coil AN .

$I_A = I_{AN}$: line current equals coil current.

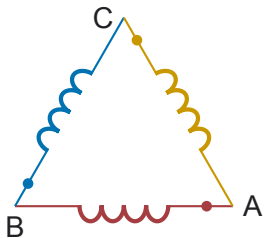
$$V_{AB} = V_{AN} - V_{BN}$$

$$V_{AB} = \sqrt{3}e^{j30^\circ} V_{AN}$$

A star point exists; neutral access requires a brought-out lead.

Line voltage is $\sqrt{3}$ times phase-coil voltage and leads it by 30° .

Δ : line and coil quantities



Balanced positive sequence

Coil AB sees line voltage V_{AB} .

I_A enters A; $I_{AB} : A \rightarrow B$;

$I_{CA} : C \rightarrow A$.

$$I_A = I_{AB} - I_{CA}$$

$$I_A = \sqrt{3}e^{-j30^\circ} I_{AB}$$

The closed delta has no neutral terminal.

Line current is $\sqrt{3}$ times coil current and lags it by 30° for the stated references.

From coils to bank terminals

Paired winding voltages

$$a = \frac{N_H}{N_L} = \frac{|E_H|}{|E_L|}$$

a compares the magnetically paired coils.

Bank terminal voltages

$$\frac{|V_{LL,H}|}{|V_{LL,L}|}$$

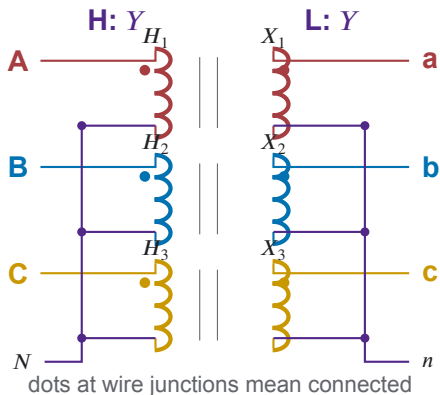
LL : line-to-line. The terminal ratio compares H/L line-voltage magnitudes.

Use one procedure for every connection

Trace the paired coil endpoints. Apply the Y/ Δ line-to-coil relations on each side. Then compare corresponding line-voltage magnitudes and angles.

First establish the coil joining; then derive what the bank terminals see.

Y–Y: physical connection



Trace the coil endpoints

The H coil ends join at N ; the L coil ends join at n .

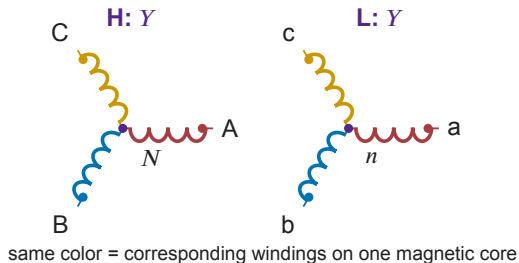
Matching pairs: $AN/an, BN/bn, CN/cn$.

Terminal marks: H_1, H_2, H_3 on H;
 X_1, X_2, X_3 on L.

X is the low-side terminal prefix.

Both winding sets have a star point.

Y–Y: terminal relations



Apply the Y relations on both sides

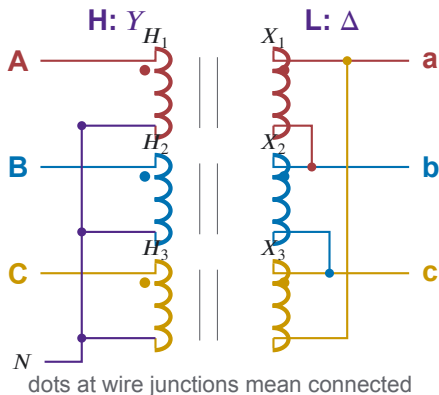
$$\frac{|V_{LL,H}|}{|V_{LL,L}|} = \frac{\sqrt{3}|E_H|}{\sqrt{3}|E_L|} = a$$

$$\frac{|I_{L,H}|}{|I_{L,L}|} = \frac{1}{a}$$

$$V_{ab,L} = V_{AB,H}/a, \quad \delta_{LH} = 0^\circ$$

The same Y factors cancel in both magnitude and phase.

Y- Δ : physical connection



Trace the coil endpoints

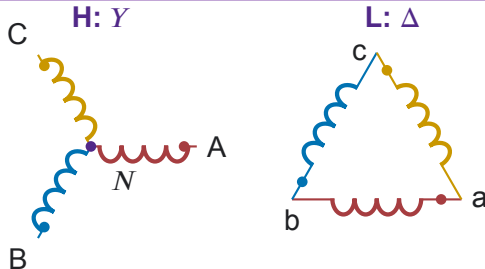
H coils: AN , BN , CN .

Matching L coils: ab , bc , ca .

Follow the undotted L ends to see how the three coil circuits close the delta.

H forms a star; L closes a delta.

Y-Δ: terminal relations

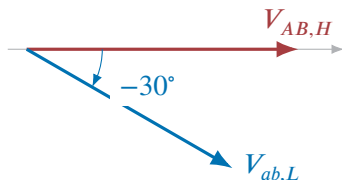


same color = corresponding windings on one magnetic core

$$\frac{|V_{LL,H}|}{|V_{LL,L}|} = \frac{\sqrt{3}|E_H|}{|E_L|} = \sqrt{3}a, \quad \frac{|I_{L,H}|}{|I_{L,L}|} = \frac{1}{\sqrt{3}a}$$

The H side contributes the Y line-voltage factor; the L side uses coil voltage directly.

Y-Δ: terminal phase shift



Use $\angle V_{AB,H} = 0^\circ$ as reference.

1. H coil AN (Y):

$$E_H = V_{AN,H} = |E_H| \angle (-30^\circ)$$

2. Transfer to paired L coil ab :

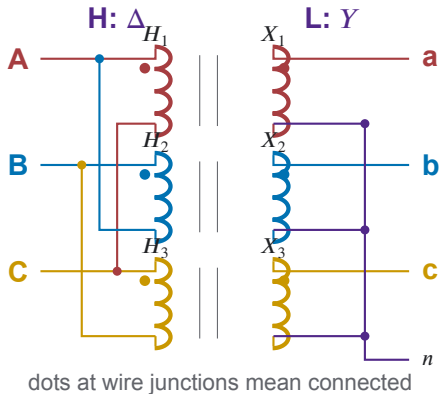
$$E_L = \frac{E_H}{a} = V_{ab,L}$$

3. Form the H line voltage (Y):

$$V_{AB,H} = \sqrt{3}e^{j30^\circ} E_H$$
$$\delta_{LH} = -30^\circ - 0^\circ = -30^\circ$$

The H line voltage leads its coil voltage by 30° ; the paired L coil is line ab .

Δ -Y: physical connection



Trace the coil endpoints

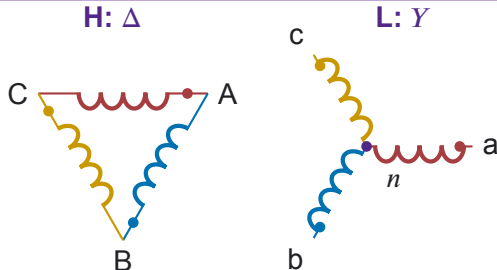
H coils: AC , BA , CB .

Matching L coils: an , bn , cn .

The L coil ends meet at n . Retain the H endpoint order when reducing this diagram.

H closes a delta; L forms a star.

Δ -Y: terminal relations

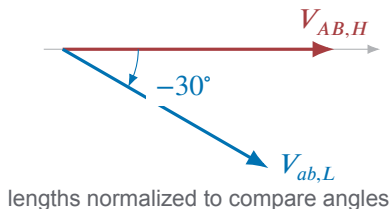


same color = corresponding windings on one magnetic core

$$\frac{|V_{LL,H}|}{|V_{LL,L}|} = \frac{|E_H|}{\sqrt{3}|E_L|} = \frac{a}{\sqrt{3}}, \quad \frac{|I_{L,H}|}{|I_{L,L}|} = \frac{\sqrt{3}}{a}$$

The L side contributes the Y line-voltage factor in the denominator.

Δ -Y: terminal phase shift



Use $\angle V_{AB,H} = 0^\circ$ as reference.

1. H coil AC (Δ):

$$E_H = V_{AC,H} = |E_H| \angle (-60^\circ)$$

2. Transfer to paired L coil an :

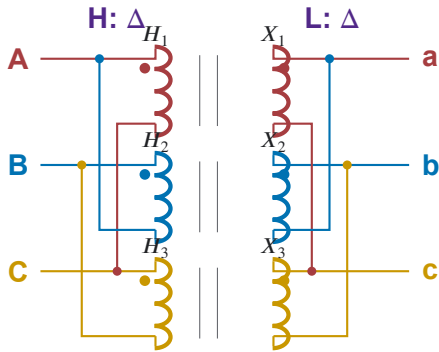
$$E_L = \frac{E_H}{a} = V_{an,L}$$

3. Form the L line voltage (Y):

$$V_{ab,L} = \sqrt{3} e^{j30^\circ} E_L$$
$$\delta_{LH} = (-60^\circ + 30^\circ) - 0^\circ = -30^\circ$$

Transfer the AC coil voltage first, then form the L line voltage.

Δ – Δ : physical connection



dots at wire junctions mean connected

Trace the coil endpoints

H coils: AC , BA , CB .

Matching L coils: ac , ba , cb .

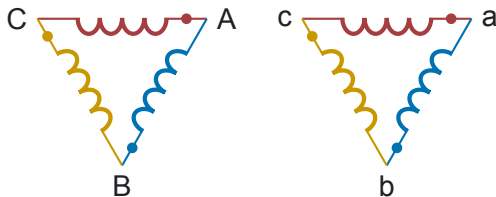
Both sides close a delta with corresponding dotted ends and the same endpoint order.

Both sides use the same cyclic joining.

Δ – Δ : terminal relations

H: Δ

L: Δ



same color = corresponding windings on one magnetic core

$$\frac{|V_{LL,H}|}{|V_{LL,L}|} = \frac{|E_H|}{|E_L|} = a, \quad \frac{|I_{L,H}|}{|I_{L,L}|} = \frac{1}{a}, \quad \delta_{LH} = 0^\circ$$

Matching delta connections preserve the paired-coil ratio and angle.

Four connections: summary

Textbook / North American standard connections; positive sequence.

H-L	H Δ order	L Δ order	$ V_{LL,H} / V_{LL,L} $	$ I_{L,H} / I_{L,L} $	δ_{LH}
Y-Y	—	—	a	$1/a$	0°
Y- Δ	—	abc	$\sqrt{3}a$	$1/(\sqrt{3}a)$	-30°
Δ -Y	ACB	—	$a/\sqrt{3}$	$\sqrt{3}/a$	-30°
Δ - Δ	ACB	acb	a	$1/a$	0°

Coil endpoints (A, B, C): $ABC = (AB, BC, CA)$; $ACB = (AC, BA, CB)$.

$$\frac{|V_{LL,H}|}{|V_{LL,L}|} \frac{|I_{L,H}|}{|I_{L,L}|} = 1, \quad S_{3\phi} = \sqrt{3}V_{LL}I_L$$

Both mixed connections use H leading L by 30° : $\delta_{LH} = -30^\circ$.

Industry notation: vector groups

Read Dyn1

Dyn1

D: H winding is delta.

y: L winding is wye.

n: L neutral terminal is brought out.

1: L lags H by 30°.

L: 11 H: 12 L: 1



Each clock hour = 30° lag.

Examples in IEC notation

Group	H-L	δ_{LH}
Dyn1	$\Delta-Y$	-30°
Dyn11	$\Delta-Y$	$+30^\circ$
Yd1	$Y-\Delta$	-30°
Dd0	$\Delta-\Delta$	0°

11 means 330° lag, equivalent to 30° lead.

n identifies neutral access; specify grounding separately.

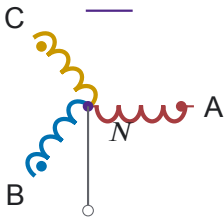
IEC 60076-1: connection symbols.

IEEE C57.12.70: terminal markings and connections.

Specify winding connections, neutral availability, and phase displacement together.

Y: neutral grounding

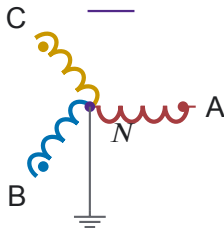
Ungrounded neutral



The star point N exists.
The neutral is intentionally ungrounded.

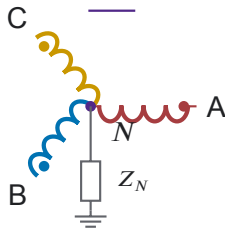
Z_N (Z_n for a low-side n): neutral grounding impedance.

Solid grounding



N is bonded to ground.
No added neutral grounding impedance.

Impedance grounding



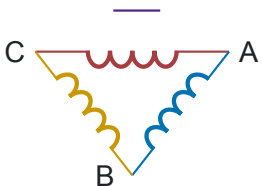
Z_N connects N to ground.
The impedance limits ground-fault current.

Y defines a star point; its neutral branch defines the grounding arrangement.

Source: Glover et al., sixth edition (2017), Fig. 3.16; Eaton, Power Distribution Systems, Grounding

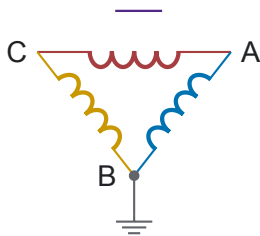
Δ : grounding choices

Ungrounded delta



No star-point neutral.
No intentional ground connection.

Corner-grounded delta



Phase *B* is bonded to ground.
B remains a phase conductor, not a neutral.

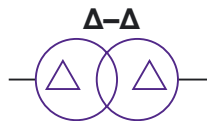
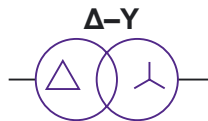
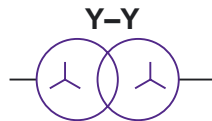
A separate grounding transformer can derive a neutral for a delta system.

A delta can be grounded, but it has no inherent star-point neutral.

Source: Schneider Electric, FAQ FA123842; Eaton, Power Distribution Systems, Grounding

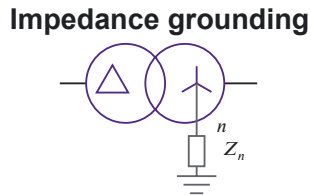
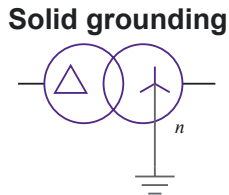
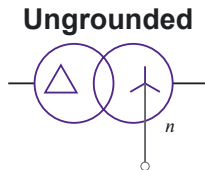
Single-line symbols and grounding

Connection symbols inside the circles (H left; L right)



Grounding is not specified in this row.

Grounding branch: the same Δ-Y connection



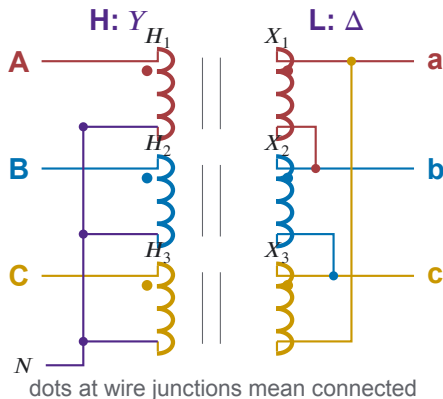
Y/Δ inside the circles; grounding on a separate branch.

Source: IEEE Std 315-1975, Secs. 6.4.15.1 and 1.9.7.1 (course-redrawn)

PART 3

Example: a three-phase transformer

Example: given data



Given: a three-phase transformer

Rated $S_{3\phi} = 60$ MVA.

Rated phase-coil voltages:

$$|E_H| = \frac{138}{\sqrt{3}} = 79.6743 \text{ kV}$$

$$|E_L| = \frac{13.8}{\sqrt{3}} = 7.96743 \text{ kV}$$

Ideal, balanced positive sequence.

At rated winding voltages.

Angle reference: $\angle V_{AB,H} = 0^\circ$.

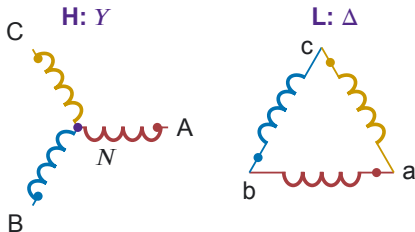
Given coil ratings and connections; find the three-phase terminal quantities.

Example: find and verify

- a** Find the winding turns ratio a , the apparent-power rating per phase, both rated line-voltage magnitudes, and their H/L quotient.
- b** Find the low-side line-voltage phasor $V_{ab,L}$ and the displacement δ_{LH} , using $\angle V_{AB,H} = 0^\circ$.
- c** Find the rated line-current magnitudes on both sides. Verify $S_{3\phi} = \sqrt{3}|V_{LL}||I_L|$ independently at H and L.
- d** Reconnect the same coils Y–Y, Δ –Y, and Δ – Δ using the Part 2 terminal conventions. Find both line voltages, both line currents, and δ_{LH} for each.

Use coil relations first; check the terminal results with the three-phase rating.

Solution (a): ratings and line voltages



same color = corresponding windings on one magnetic core

$$a = 79.6743/7.96743 = 10$$

$$S_{3\phi}/3 = 60/3 = 20 \text{ MVA}$$

Apply one line-to-coil rule per side

$$|V_{LL,H}| = \sqrt{3}|E_H| = 138 \text{ kV}$$

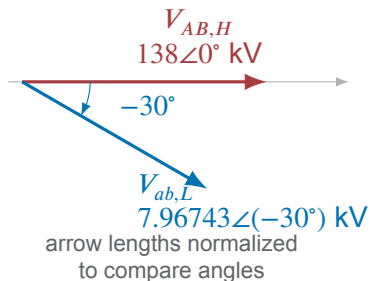
$$|V_{LL,L}| = |E_L| = 7.96743 \text{ kV}$$

$$\frac{|V_{LL,H}|}{|V_{LL,L}|} = \frac{138}{7.96743} = 17.3205$$

The winding turns ratio is still $a = 10$.

The Y-side line-voltage factor turns a coil ratio of 10 into a terminal ratio of 17.3205.

Solution (b): voltage phasors



Trace the red coil pair

$$V_{AB,H} = 138\angle 0^\circ \text{ kV}$$

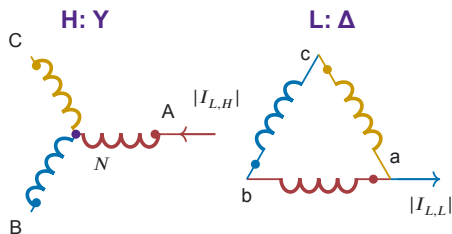
$$V_{AN,H} = \frac{138}{\sqrt{3}}\angle(-30^\circ) \text{ kV}$$

$$V_{ab,L} = \frac{V_{AN,H}}{10}$$

$$\delta_{LH} = -30^\circ - 0^\circ = -30^\circ$$

The transformer preserves the paired coil angle; forming the H line voltage creates the 30° displacement.

Solution (c): rated currents



$$S_{\phi} = \frac{60}{3} = 20 \text{ MVA}$$

$$|I_{\text{coil}}| = \frac{S_{\phi}}{|E|}$$

From coil ratings to terminal ratings

	H (Y)	L (Δ)
Coil voltage (kV)	79.6743	7.96743
Coil current (A)	251.022	2510.22
Line/coil current factor	1	$\sqrt{3}$
Line current (A)	251.022	4347.83
Line voltage (kV)	138	7.96743
Total rating (MVA)	60	60

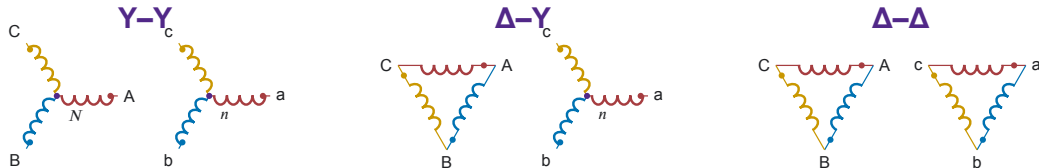
$$S_{3\phi} = \sqrt{3}|V_{LL}||I_L|.$$

Using kV and kA gives MVA on each side.

$17.3205 \times 0.057735 \approx 1$: the terminal voltage and current factors compensate.

Solution (d): reconnect the coils

Keep the same coil ratings: $a = 10$, $S_{3\phi} = 60$ MVA.



Connection	$ V_{LL,H} $ (kV)	$ V_{LL,L} $ (kV)	$ I_{L,H} $ (A)	$ I_{L,L} $ (A)	δ_{LH}
Y-Y	138	13.8	251.022	2510.22	0°
Δ -Y	79.6743	13.8	434.783	2510.22	-30°
Δ - Δ	79.6743	7.96743	434.783	4347.83	0°

Same coil ratings; new terminal values; both sides retain 60 MVA.

Summary

- 1 Identify the winding sets, phase system, and physical construction.
- 2 Trace matching coil endpoints before reducing the connection diagram.
- 3 Apply Y/ Δ voltage and current factors: the example has $a = 10$ but a Y- Δ terminal voltage ratio of 17.3205.
- 4 Use the terminal phasors to obtain $\delta_{LH} = -30^\circ$; verify the 60 MVA rating on both sides.

Balanced bank models, autotransformers, and taps use these winding and terminal relations.